Unit: 6 Unit 2: Exponents Radicals
Topic: Rationalizing Denominator

Name: ________________
Date: ___________

Advanced (Habanero)

Q1. Rationalize the denominator of $\frac{1}{\sqrt{2}}$

(A) $\frac{\sqrt{2}}{2}$
(B) $\frac{1}{2\sqrt{2}}$
(C) $\frac{2}{\sqrt{2}}$
(D) $\frac{\sqrt{2}}{4}$
(E) $\frac{1}{4\sqrt{2}}$

Q2. Simplify $\frac{1}{\sqrt{3}+\sqrt{5}}$

(A) $\frac{\sqrt{3}-\sqrt{5}}{-2}$
(B) $\frac{\sqrt{5}-\sqrt{3}}{2}$
(C) $\frac{\sqrt{3}+\sqrt{5}}{2}$
(D) $\frac{\sqrt{3}-\sqrt{5}}{2}$
(E) $\frac{\sqrt{5}+\sqrt{3}}{-2}$

Q3. Rationalize the denominator of $\frac{2}{\sqrt{6}-\sqrt{3}}$

(A) $\frac{2(\sqrt{6}+\sqrt{3})}{3}$
(B) $\frac{2(\sqrt{6}-\sqrt{3})}{9}$
(C) $\frac{2(\sqrt{3}-\sqrt{6})}{-3}$
(D) $\frac{2(\sqrt{6}+\sqrt{3})}{9}$
(E) $\frac{2(\sqrt{3}+\sqrt{6})}{3}$

Q4. Simplify $\frac{\sqrt{2}+1}{\sqrt{2}-1}$

(A) $3+2\sqrt{2}$
(B) $3-2\sqrt{2}$
(C) $2+\sqrt{2}$
(D) $\sqrt{2}+1$
(E) $\sqrt{2}-1$

Q5. Rationalize the denominator of $\frac{1}{\sqrt{7}+\sqrt{5}}$

(A) $\frac{\sqrt{7}-\sqrt{5}}{2}$
(B) $\frac{\sqrt{5}-\sqrt{7}}{2}$
(C) $\frac{\sqrt{7}+\sqrt{5}}{2}$
(D) $\frac{\sqrt{5}+\sqrt{7}}{-2}$
(E) $\frac{\sqrt{7}-\sqrt{5}}{-2}$

Q6. Simplify $\frac{\sqrt{3}}{\sqrt{6}+1}$

(A) $\frac{\sqrt{3}(\sqrt{6}-1)}{5}$
(B) $\frac{\sqrt{3}(\sqrt{6}+1)}{7}$
(C) $\frac{\sqrt{6}-1}{\sqrt{3}}$
(D) $\frac{\sqrt{6}+1}{\sqrt{3}}$
(E) $\frac{\sqrt{6}-1}{\sqrt{2}}$

Q7. Rationalize the denominator of $\frac{1}{\sqrt{10}-\sqrt{2}}$

(A) $\frac{\sqrt{10}+\sqrt{2}}{8}$
(B) $\frac{\sqrt{10}-\sqrt{2}}{-8}$
(C) $\frac{\sqrt{10}-\sqrt{2}}{8}$
(D) $\frac{\sqrt{2}+\sqrt{10}}{-8}$
(E) $\frac{\sqrt{2}-\sqrt{10}}{8}$

Q8. Simplify $\frac{\sqrt{5}}{\sqrt{10}-\sqrt{2}}$

(A) $\frac{\sqrt{5}(\sqrt{10}+\sqrt{2})}{8}$
(B) $\frac{\sqrt{10}-\sqrt{2}}{\sqrt{5}}$
(C) $\frac{\sqrt{2}+\sqrt{10}}{\sqrt{5}}$
(D) $\frac{\sqrt{10}+\sqrt{2}}{\sqrt{5}}$
(E) $\frac{\sqrt{5}(\sqrt{2}+\sqrt{10})}{-8}$

Open Ended Questions (FRQ)

Q9. Rationalize the denominator and simplify: $\frac{2\sqrt{3}+1}{\sqrt{6}-1}$

Q10. Rationalize the denominator and simplify: $\frac{\sqrt{2}+\sqrt{3}}{\sqrt{6}-\sqrt{2}}$

Chef's Tips

- Make sure to simplify all radicals
- Check for any remaining radicals in the denominator
- Use the conjugate to rationalize the denominator